

Demonstration of Proposed Features

Date	12 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S. No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Applicant Data Intake Form	Web form (templates/index.html) capturing personal, financial, and employment details.	Implemented	Yes	Covers age, gender, income, education, occupation, housing type, and more.
2	Feature Engineering Pipeline	Converts raw form input into the exact one-hot-encoded feature vector the model expects (build_feature_vector).	Implemented	Yes	Includes derived fields such as DEPENDENT_S_RATIO and NOT_EMPLOYED_FLAG.
3	Vintage-Filtered Label Engineering	Labels applicants using ≥ 24 months of observed credit history to reduce label noise (Epic 3, v3).	Implemented	Yes	Produced a +19% relative F1 improvement over the unfiltered (v2) label.
4	Tuned XGBoost Prediction Model	Cross-validated, hyperparameter-tuned XGBoost classifier with a custom decision threshold (0.29).	Implemented	Yes	Test metrics: precision (reject) $\approx$ 41.8%, recall (reject) $\approx$ 40.2%, F1 $\approx$ 41.0%.
5	Real-Time Prediction Endpoint	/predict Flask route that returns an Approve/Reject decision with approval probability.	Implemented	Yes	Responds within seconds of form submission.

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6	Result Visualization & Disclosure	Probability gauge (SVG) plus a plain-language panel disclosing model precision/recall limitations.	Implemented	Yes	Ensures predictions are not presented as more reliable than they are.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%